

SOUHIL SID

Senior XR & Digital Twins Technologist

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XR, digital-twin and AI technologist with 6+ years of hands-on experience building immersive simulations, serious games, computer-vision interactions, digital laboratories, robotic interfaces and engineering visualization systems. Combines Unity/C#, AR/VR/MR, 3D medical data pipelines, Python/ML, Bayesian optimization, robotics and real-time sensing to turn complex scientific and clinical workflows into interactive, measurable experiences. Experienced in rapid prototyping, technical R&D, experimental validation, stakeholder demonstrations and full-cycle delivery from concept and data preparation through deployment-ready builds.

PROFESSIONAL EXPERIENCE

AISCIA Informatics | R&D Engineer - AI, Digital Twins & Laboratory Automation

Sep 2024 - Present

Doha, Qatar

- Lead applied R&D combining AI, optimization, simulation, digital twins, robotics and interactive technical interfaces for accelerated experimentation and scientific decision support.
- Designed and delivered a self-driving laboratory integrating robotic liquid handling, automated dosing/mixing, software control, experimental data capture, digital twins and AI-guided closed-loop optimization.
- Build automated engineering evaluators linking simulator or real-time data to predictive models, What-If analysis and optimization recommendations.
- Translate research concepts into usable products through requirements, architecture, prototyping, validation, QA, documentation, demonstrations and cross-functional handoff.

Qatar Skills Training Center | Unity Developer - Serious Games & Interactive Training (Part Time)

August 2026

Doha, Qatar

- Develop Unity-based simulations, serious-game mechanics and interactive learning experiences for professional training, with emphasis on engagement, practical skill-building and measurable outcomes.
- Create rapid prototypes, 3D interaction flows and reusable training systems suitable for workshops, demonstrations and facilitator-led programs.

Descifer Agency | Unity & Computer Vision Developer / Phygitl Game Developer

Project Contract

Doha, Qatar

- Delivered six production-ready phygitl games in approximately six weeks using Unity and camera-based computer vision for real-world interactive installations.
- Built camera interaction, calibration workflows, gameplay loops, scoring, collision/trigger logic, user feedback and deployment flows for public-facing experiences.
- Handled fast iteration from concept and mechanics through debugging, optimization and final playable delivery under compressed production timelines.

Develop Together Corp | Founder, CEO & Game / XR Developer

Feb 2020 - Present

Boumerdes, Algeria / Remote

- Lead end-to-end production of educational games, AR/VR applications, simulations, digital twins and interactive technology prototypes.
- Develop concepts, GDDs, technical architecture, Unity/C# systems, source-code delivery, client communication, mentoring and technical direction.
- Published and delivered award-winning mobile games, VR learning experiences, digital-twin concepts and industrial simulation prototypes.

World Learning Algeria | Game Development & VR Intern

Feb 2023 - Feb 2024

STEM Mentor

Algiers, Algeria

- Developed a virtual twin of a STEM Makerspace with immersive activities spanning robotics, games, physics and astronomy.
- Managed Tales of Knowledge, an educational cultural game covering storyline, GDD, mechanics, animation and learner interaction.
- Delivered STEM and immersive-learning activities using gamification, Universal Design for Learning and project-based learning.

Shedio Design Solutions | Game & VR Developer (Part Time)

Mar 2020 - Feb 2023

Hassi R'Mel, Algeria

- Created VR and interactive solutions including a United Nations virtual tour, virtual art gallery and oil-and-gas industrial training simulations.
- Supported mentoring and IT activities while developing Unity-based immersive visualization and training products.

FLAGSHIP XR & DIGITAL TWIN PROJECTS

Emergency Cerebral Aneurysm Digital Twin for Microcatheter Navigation [Medical R&D Prototype]

Unity • SimVascular • Bayesian Optimization • Python • Patient-Specific 3D Geometry • Hemodynamic / Navigation What-If Scenarios

- Designed an R&D digital-twin platform concept for patient-specific cerebral aneurysm intervention planning, combining vascular geometry, simulation outputs and an interactive Unity environment.
- Structured a workflow in which anatomy and flow-related simulation data can be explored in 3D while candidate navigation or procedural parameters are evaluated using Bayesian optimization rather than exhaustive trial-and-error.
- Focused on emergency-use research scenarios: vessel exploration, microcatheter-path visualization, scenario comparison, quantitative feedback and optimization-assisted planning, explicitly as a prototype/research system rather than a clinical decision device.

VR Liver Surgery Simulation & Immersive Anatomy Training [Medical VR]

Unity • VR • Oculus / OpenXR • 3D Anatomy • Interaction Design • Procedural Training

- Developed an immersive liver-surgery simulation concept for anatomy orientation and procedural rehearsal, translating 3D anatomical structures into an interactive VR environment.

- Implemented spatial inspection, object manipulation, viewpoint control, tool interaction and step-based training concepts suitable for teaching and instructor-guided demonstrations.
- Structured the experience as reusable modules so anatomy, instruments, scenarios, prompts and assessment logic can evolve without rebuilding the application.

MRI / MRA Visualization and Simulation Pipeline [Medical Imaging]

3D Slicer • DICOM / MRI / MRA • Segmentation • 3D Reconstruction • Unity • Interactive Visualization

- Created a workflow for converting MRI/MRA studies into segmented 3D structures with 3D Slicer, then importing optimized geometry into Unity for interactive visualization and simulation prototyping.
- Explored vessel/anatomy isolation, mesh cleanup, scale/orientation consistency, labeling, clipping/layered visualization and case-specific scene organization.
- Designed the pipeline for future extension toward patient-specific XR, measurements, collaborative review, procedural simulation and digital-twin applications.

Digital Twin-Integrated Robotic Platform for Autonomous Liquid Handling [Robotics + AI]

Unity / Digital Twin • Robotics • AI-Assisted Experimentation • Automated Dosing • Data Capture • Closed-Loop Optimization

- Designed and delivered a self-driving laboratory platform from research through production, integrating robotic liquid handling, automated dosing/mixing, software control, digital-twin visualization and experimental data capture.
- Connected experimental workflows to AI-guided optimization so each experiment could inform the next candidate condition, enabling closed-loop R&D and reducing manual iteration.
- Presented the platform in technical demonstrations and international innovation settings; related work was accepted as an AI4X-AC 2026 poster on scalable, low-cost laboratory automation.

Six Phygital Computer-Vision Games for Qatar Activations [Computer Vision]

Unity • Computer Vision • Camera Interaction • Calibration • Real-Time Scoring • Physical-Digital Gameplay

- Delivered six camera-driven phygital games on a rapid production schedule, combining physical movement with real-time Unity gameplay.
- Implemented calibration, player-zone logic, collision/trigger mapping, scoring, feedback and installation flows designed for repeated public use.
- Tuned performance and interaction behavior for variable lighting, player movement and physical-space constraints.

Kinect-Based Motion Interaction & Touchless Training Prototypes [Kinect / Spatial Computing]

Microsoft Kinect / Depth Sensing • Skeleton Tracking • Gesture Interaction • Unity • Spatial Mapping • HCI

- Built motion-driven interaction prototypes using Kinect/depth-sensing input, mapping body movement and gestures to real-time Unity systems.
- Developed touchless selection, gesture-triggered actions, movement tracking and spatial-feedback patterns suited to training, rehabilitation-style interaction, exhibitions and hands-free environments.
- Handled coordinate mapping and calibration between the physical tracking space and virtual scene to produce stable, intuitive interactions.

Oculus / Meta Quest & OpenXR Immersive Training [VR]

Meta Quest / Oculus • OpenXR • XR Interaction Toolkit • Hand / Controller Interaction • Spatial UI • VR Locomotion

- Built immersive training and simulation prototypes using Unity XR workflows, controller interaction, teleportation/locomotion, object manipulation and spatial UI.
- Designed reusable interaction prefabs and scenario logic for industrial, STEM and medical-style experiences, supporting rapid adaptation across domains.
- Optimized scene complexity, feedback and user flow for headset-based deployment concepts.

Predictive Digital Twin of a Heat Exchanger [Real-Time Twin]

Unity • LabVIEW • Arduino • Python • Real-Time Monitoring • Predictive Analytics • What-If Scenarios

- Integrated sensor/experimental data with Unity visualization and engineering calculations for heat duty, overall heat-transfer coefficient and LMTD.
- Connected the twin to predictive models and What-If scenarios to forecast performance, identify abnormal behavior and communicate operational recommendations visually.
- Demonstrated a full digital-twin architecture combining physical measurements, mathematical models, predictive intelligence and an interactive 3D interface.

Industrial Process & Oil/Gas Immersive Simulators [Industrial XR]

Unity • VR • 3D Industrial Equipment • Turbines • Compressors • Refining • Training Simulation

- Created interactive 3D/VR simulations for turbines, compressors, refining systems and process-engineering concepts, translating complex equipment behavior into accessible training experiences.
- Combined engineering understanding with interaction design so users could inspect equipment, understand flows and operating logic, and engage with scenario-based training.

TECHNICAL TECHNOLOGY STACK

Unity / XR	Unity3D, C#, OpenXR, XR Interaction Toolkit, Oculus / Meta Quest, HTC VIVE, AR/VR/MR, physics, UI, animation, mobile/standalone builds, serious games and simulation.
Medical 3D	3D Slicer, DICOM-oriented workflows, MRI/MRA visualization, segmentation, 3D reconstruction, anatomy mesh preparation, spatial labeling and Unity integration.
Digital Twins	Real-time/offline twins, What-If scenarios, sensor and simulator data integration, predictive analytics, scientific visualization and interactive technical dashboards.
AI / Optimization	Python, machine learning, Bayesian optimization, active learning, DOE/RSM, inverse design, multi-objective/Pareto optimization and automated evaluators.
Computer Vision	OpenCV prototyping, camera-based interaction, calibration, phygital gameplay, body-driven input and Kinect/depth-sensing interaction.
Robotics / IoT	Robotic liquid handling, Arduino, LabVIEW, sensor integration, automated dosing/mixing, IoT concepts and closed-loop laboratory workflows.
Engineering Sims	COMSOL, Aspen HYSYS, Aspen Plus, DWSIM, GPS-X, CapdetWorks, HOMER Pro, System Advisor Model and AutoCAD Plant 3D.
Delivery	R&D scoping, literature/technology review, requirements, GDDs, rapid prototyping, QA/testing, validation planning, documentation, demonstrations and stakeholder communication.

SELECTED RESEARCH, INNOVATION & TECHNICAL LEADERSHIP

- Accepted poster, AI4X-AC 2026 - "Scalable Low-Cost Laboratory Automation: A Digital Twin-Integrated Robotic Platform for Autonomous Liquid Handling."
- Founder of an XR Lab initiative at the University of Boumerdes to help faculty integrate immersive technologies into engineering and science education.
- President, AIChE Student Chapter, University of Boumerdes (2022-2023) - led technical training, engineering project teams and immersive process-simulation initiatives.
- Represented AISCIA at multinational conferences and innovation forums in Qatar and internationally, presenting AI-enabled materials discovery, laboratory automation and sustainable-engineering solutions.
- World Learning Algeria STEAM mentor and volunteer (2021-2025) - delivered robotics, engineering, game-development and project-based learning activities.

AWARDS & RECOGNITION

- 1st Place - Xylem Global Water Innovation Challenge, 2024
- 1st Place - Game Development World Championship, Mobile Category, 2023
- 5th Place - EAGE CO₂ Minus Challenge, 2024
- Top 3 Winner - JunctionX International Hackathon, 2023
- NASA Space Apps - Galactic Problem Solver, 2022
- International exhibitions & technology demonstrations in Qatar and abroad

EDUCATION & CERTIFICATIONS

Master of Process Engineering - Petroleum Refining

Faculty of Hydrocarbons and Chemistry, University of Boumerdes, Algeria | 2022-2024 | GPA 13.94/20

- Advanced engineering foundation supporting scientific simulation, fluid/process understanding, modeling, thermodynamics, process safety and multidisciplinary digital-twin development.

Bachelor of Process Engineering

University of Boumerdes, Algeria | 2019-2022 | GPA 13.13/20

- Foundation in transport phenomena, heat transfer, kinetics, thermodynamics, chemistry, process simulation and engineering analysis.

Unity VR Development Pathway - Unity Technologies, 2024: Unity3D, XR Management, OpenXR, HTC VIVE, Oculus, XR Interaction Toolkit, digital twins and passthrough.

Unity Junior Programmer Pathway - Unity Technologies, 2021: C#, object-oriented programming, game physics, save/load systems, multiplayer, game UI and simulations.

STEM Teacher Training - World Learning Algeria, 2020: Universal Design for Learning, gamification, lesson planning, game-development tools and SMART goals.

LANGUAGES

Arabic - Native | **English** - Advanced | **French** - Intermediate